

“Openness Challenges The Traditional Closed Nature Of Hardware IPs To Create Standardised Solutions”

In an interview with EFY's Yashasvini Razdan, Alex P. James, Dean of Academics at Digital University Kerala, argues that an open AI hardware ecosystem could democratise access to hardware design tools and IPs, much like the Linux movement did for software.

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What do you mean by an open AI hardware ecosystem?

In hardware design, intellectual properties (IPs) are often closed, posing a challenge for countries like India with abundant talent but limited access to expensive tools. Efforts such as Chip to Startup aim to democratise access, but they reach only a fraction of potential users, considering the vast student base and aspiring researchers. Therefore, the concept of openness entails two crucial aspects. First, openness provides access to free and open source tools, thus expanding accessibility beyond those restricted by costly proprietary software. Second, it pertains to mak-

ing IPs open, enabling democratised development where individuals from diverse backgrounds can contribute, akin to the Linux open source movement. Embracing openness allows for an open innovation culture in hardware. Openness challenges the traditional closed nature of hardware IPs to create standardised solutions, such as a universally open USB standard, or other such technologies.

How will it affect business opportunities in the electronics hardware ecosystem?

It is only going to increase. So, keeping things open is going to reduce the barrier of entry. Currently,

funding and access to tools and specific IPs are major barriers. With an open framework, access to these tools, IPs, and design methodologies becomes much easier, allowing more people to start. While many first-year students can easily use Python libraries to build applications, far fewer can build a chip or have access to the necessary resources.

What about AI tools which can help designers with entire chip designs?

True, but how do you validate it against a problem? If there is an IP owned by another entity and the designer tries to generate it, they will run into legal complications.